

CO-2 AT-1 Analytical Problem Solving

Name: H. Harshavardhan

Regno: 192S11171

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Sub: CSA0315 Data structure

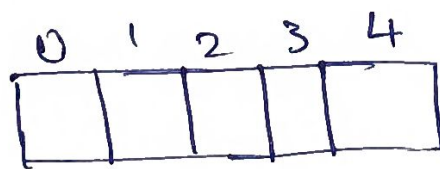
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- 1) Perform the following operations in LIFO data structure having an array size of 5, Push (45), Push (22), Push (11), Push (77), POP(), Push (44), Push (33), POP(), POP(). Print the peak element of the data structure and its index position with a neat sketch explain in detail.

Sol

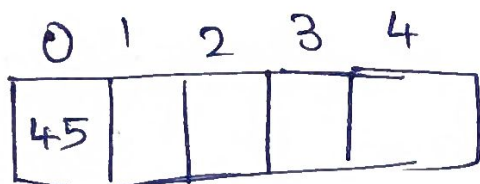
Size of stack = 5

Initially top = -1



(i) Push (45)

top = 0

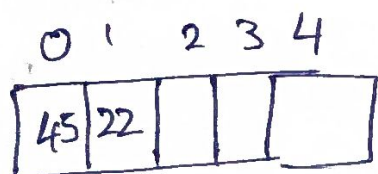


(ii) Push (22)

top = top + 1

= 0 + 1

= 1



(iiz)

Push (11)

0	1	2	3	4
45	22	11		

$$\text{top} = \text{top} + 1$$

$$= 1 + 1$$

$$= 2$$

iv)

Push (77)

0	1	2	3	4
45	22	11	77	

$$\text{top} = \text{top} + 1$$

$$= 2 + 1$$

$$= 3$$

v)

Pop()

0	1	2	3	4
45	22	11		

$$\text{top} = \text{top} - 1$$

$$= 3 - 1$$

$$= 2$$

vi)

Push (44)

0	1	2	3	4
45	22	11	44	

$$\text{top} = \text{top} + 1$$

$$= 2 + 1$$

$$= 3$$

vii)

Push (33)

0	1	2	3	4
45	22	11	44	33

$$\text{top} = \text{top} + 1$$

$$= 3 + 1$$

$$= 4$$

viii)

Pop()

45	22	11	44	
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$$\text{top} = \text{top} - 1$$

$$= 4 - 1$$

$$= 3$$

Pop()

0	1	2	3	4
45	22	11		

$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

Peak element of the stack is 22 in position 1
 $\text{top} = \text{top} - 1 = 2 - 1 = 1$

0	1	2	3	4
45	22			

2) Perform the following operations in Circular Queue having a size of 5.
 Enqueue (20), Enqueue (10),
 Enqueue (30), Dequeue (100), Dequeue (100), Dequeue (100),
 Enqueue (90), Enqueue (100), Dequeue (1), Dequeue (1),
 Enqueue (60), Enqueue (90). Finally display the front and rear elements with its positions. Discuss in detail.

size of queue = 5

Initially Front of Rear = -1

(i) Enqueue (20)

$$\text{Front} = \text{Rear} = 0$$

0	1	2	3	4
20				

ii) Enqueue (10)

$$\text{Front} = 0, \text{Rear} = \text{Rear} + 1 = 1$$

0	1	2	3	4
20	10			

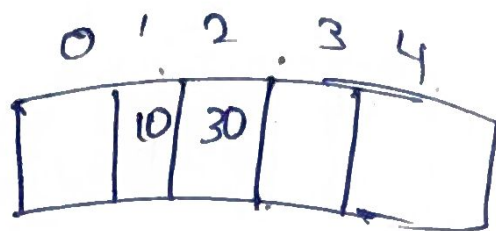
(iii) Enqueue (30)

$$\text{Front} = 0, \text{Rear} = \text{Rear} + 2 = 2$$

0	1	2	3	4
20	10	30		

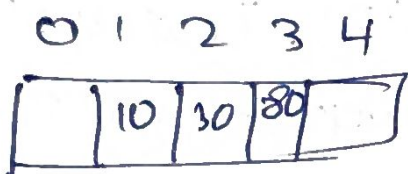
(iv) Dequeue

$$\begin{aligned}\text{Front} &= \text{Front} + 1 \\ &= 0 + 1 \\ &= 1\end{aligned}$$

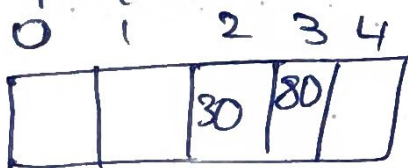


$$\begin{aligned}\text{Rear} &= \text{Rear} + 1 \\ &= 2 + 1 \\ &= 3\end{aligned}$$

(v) Enqueue (80)

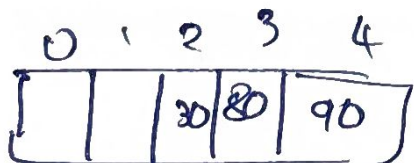


(vi) Dequeue ()



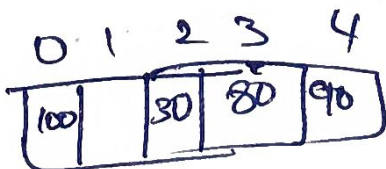
$$\begin{aligned}\text{Front} &= \text{Front} + 1 \\ &= 1 + 1 \\ &= 2\end{aligned}$$

(vii) Enqueue (90)



$$\begin{aligned}\text{Rear} &= \text{Rear} + 1 \\ &= 3 + 1 \\ &= 4\end{aligned}$$

(viii) Dequeue (100)

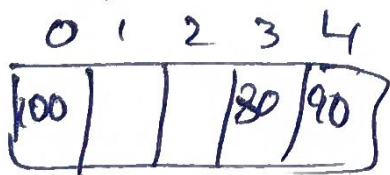


$$\begin{aligned}\text{Rear} &= \text{Rear} + 1 \\ &= 5\end{aligned}$$

The queue is full

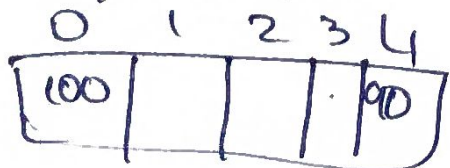
Since it is greater Rear is 0

(ix) Dequeue ()

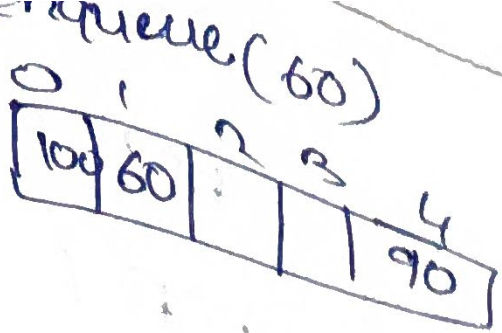


$$\begin{aligned}\text{Front} &= \text{Front} + 1 \\ &= 2 + 1 \\ &= 3\end{aligned}$$

(x) Dequeue ()



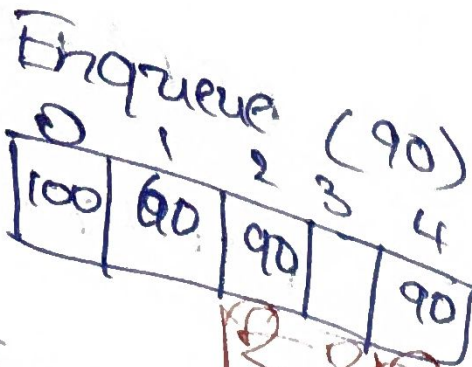
$$\begin{aligned}\text{Front} &= \text{Front} + 1 \\ &= 3 + 1 \\ &= 4\end{aligned}$$



$$\text{Rear} = \text{Rear} + 1$$

$$\text{Rear} = 0 + 1$$

$$= 1$$



$$\text{Rear} = \text{Rear} + 1$$

$$\text{Rear} = 1 + 1$$

$$= 2$$

Convert the following infix expression into a Post fix expression

$$B - [C/D + (E + F * G) / H]^J$$

Also evaluate the given postfix expressions.

a) $9 \ 3 \ 84 \ / \ ^* \ +$

b) $62 \ + \ 63 \ / \ ^*$

Scanning Character	stack	Postfix
B		B
-	-	B
[	[-	B
C	[-C	BC
/	[-C/	

BCD / FFG 7. 4H 4J*

OVER

a) 9 3 8 4 / * +

Scanning Character	Stack	Post fix
9	[9]	
3	[9 3]	
8	[9 3 8]	
4	[9 3 8 4]	
/	[9 3 2]	$8/4 = 2$
*	[9 6]	$8 \times 2 = 6$
+	[15]	$9 + 6 = 15$
Over	Stack empty	Ans = 15

b) 6 2 + 6 3 / *

Scanning Character	Stack	Post fix
6	[6]	
2	[6 2]	
+	[8]	
6	[8 6]	
3	[8 6 3]	

$6 + 2 = 8$

Sub

1

A

DVER

812

16

empty

$6/3 = 2$

$8 \times 2 = 16$

O/P = 16